PRETOM AREEFIN PRANTO

COMPUTER SCIENCE • MACHINE LEARNING • SOFTWARE ENGINEERING

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PROFESSIONAL SUMMARY

Computer Science undergraduate at BRAC University, with practical experience in machine learning engineering, multimodal AI research, data annotation, and full-stack development. Completed an internship at FlyRank AI and developed academic projects involving healthcare image classification, biological data analysis, and MERN-stack web applications.

EDUCATION

BRAC University | Bachelor in Computer Science 2022 – Jan 2027 (expected)

Relevant coursework: Machine Learning, Neural Networks, Software Engineering, Database Management, Computer Networks

TECHNICAL SKILLS

Programming: Python, JavaScript, PHP, SQL (MySQL), HTML, CSS

ML & Data: PyTorch, MONAI, scikit-learn, XGBoost, Pandas, NumPy, CNNs, ResNet, DenseNet, clustering, embeddings, feature engineering, model evaluation (ROC-AUC, F1, Precision etc.)

Software & Web: MongoDB, Express.js, React, Node.js (MERN), REST APIs

AI & Data Annotation: Text and image annotation, prompt engineering

Tools: Git, GitHub, Google Colab, VS Code, LaTeX, Canva, Spyder, Flex

RELEVANT PROFESSIONAL EXPERIENCE

Machine Learning Engineering Intern | FlyRank AI | GitHub Link Jul – Oct 2026

- Completed a Google Search Ranking & Discoverability capstone for content-opportunity scoring using approximately 30,000 Google Search Console/GA4 records; final project accepted by the lead track mentor.
- Identified an overly optimistic initial evaluation (Precision@10 = 0.70) and used client-grouped train/test splits to reduce data leakage, reporting out-of-sample Precision@10 = 0.40.
- Applied data wrangling, embeddings, clustering, and opportunity modeling; completed 118 internship program hours and 14 verified Anthropic Academy courses.

AI Data Trainer | Genmorphics AI, Bangladesh 2024

- Annotated text and image data for AI workflows and developed localized prompts for AI-related tasks.

RESEARCH EXPERIENCE

Undergraduate Thesis | BRAC University | Department of Computer Science and Engineering (2026)

NeuroVentura: A Modern Approach to Predict Alzheimer's Disease Risk and Progression Through Multimodal Machine Learning Models

Supervisor: Md. Saiful Bari Siddiqui (Senior Lecturer) **Co-supervisor:** Dr. Swakkhar Shatabda (Professor)

- Developed a late-fusion framework combining a frozen MONAI 3D DenseNet-121 MRI, XGBoost on MMSE, age, and gender, and a residual MLP on a matched cohort.

MACHINE LEARNING & AI PROJECTS

LungCare — Multiclass Lung Cancer Detection | **International Biotechnology Conference, 2025**

- Developed a deep learning pipeline for lung cancer subtype classification from histopathology images, including image preprocessing and prediction visualization.

Diabetic Retinopathy Severity Classification | **CSE427 | Machine Learning** | GitHub Link

- Developed a four-class diabetic retinopathy severity classifier using ResNet50 transfer learning, class-weighted training, and Grad-CAM.

Music Generation Using Unsupervised Learning | **CSE425 | Neural Networks** | **GitHub Link**

- Compared LSTM autoencoder, VAE, and Transformer approaches for musical sequences using MAESTRO v3.0.0 MIDI files and piano-roll representations.

Mice Protein Expression Classifier | **CSE422 | Machine Learning** | **Colab Link**

- Built Python classification workflows for mouse protein-expression data using preprocessing, feature engineering, and supervised learning algorithms.

PretBuddy — A Realtime AI Chat Voice Assistant | **Personal project** | **GitHub Link**

- A real-time Python- AI oriented voice assistant using AssemblyAI, Groq web search, and ElevenLabs.

SOFTWARE ENGINEERING PROJECTS

Abar Nosto! — Community Issue Monitoring Platform | **CSE471 | MERN** | **Live Link**

- Built and deployed a MERN platform on Vercel for community issue reporting, featuring an activity feed, reputation system, offline drafts, and administrative tools.

BRACU Student Hub | **CSE470 | Software Engineering** | **GitHub Link**

- Developed a MERN platform for academic resources, exam deadlines, calendar features, faculty reviews, and administrator-led content management.

AidTrack — Healthcare Management Platform | **CSE370 | Database Management**

- Developed a PHP, HTML, CSS, and database-backed healthcare platform for appointments, patient records, and doctor–patient communication.

Weather CLI — Python Package | **Personal project** | **GitHub Link**

- Developed a typed Python CLI with TTL caching and a rich terminal interface; added pytest tests (94% coverage) and GitHub Actions CI.

ADDITIONAL EXPERIENCE

Tech Writer & Researcher (Freelance) | **Upwork and Fiverr** 2024 – Present

- Produced research-based academic and technology articles, with supporting visual content created in Canva.

Social Media Executive | **G17 UAC Bangladesh** 2024 – 2026

- Managed social media campaigns aligned with UN Sustainable Development Goals and represented BRAC University

News Writer & Researcher | **The Papyrus | Online News Media** 2024 – 2025

- Researched and wrote technology and social-interest articles; supported search engine optimization and created editorial graphics.

Senior Martial Arts Instructor and Martial Artist | **Dragon Council Bangladesh** 2019 – 2024

- Internationally acclaimed martial artist, taught Shaolin Kung Fu to students, maintained formal regulation